

Chedam P0: Raspberry Pi hub setup on the new network

Setup log · Saturday, October 3, 2026 · Owner: Sreya · Companion to docs/P0-restart-new-network.md (the plan); this file records what was actually done.

1. Outcome

The Chedam hub (Raspberry Pi Zero 2 W) was re-flashed and is running on the new router's Wi-Fi with a reserved IP address. Login from the PC works with an SSH key only, password login over SSH is off, the OS is fully updated, and the HDMI screen shows the console login prompt.

Item	Final state
Hardware	Raspberry Pi Zero 2 W Rev 1.0 (Wi-Fi only, no Ethernet port, one mini-HDMI port)
OS	Raspberry Pi OS Lite 64-bit, Debian 13 (trixie), kernel 6.18.50+rpt-rpi-v8; no desktop (multi-user.target)
Hostname / user	chedam / chedam
Network	Own TP-Link router (behind the TELUS modem), subnet 192.168.50.0/24, gateway 192.168.50.1, 2.4 GHz Wi-Fi
Pi address	192.168.50.101 on wlan0, MAC 88:A2:9E:5C:17:65, DHCP reservation enabled
Dev PC	DESKTOP-AHG4TD7 (Windows), 192.168.50.100
SSH	Key login with id_ed25519 (comment chedam-dev), public key in /home/chedam/.ssh/authorized_keys on the Pi; passwordauthentication no; alias ssh chedam
Display	Console login visible on HDMI after forcing HDMI output in cmdline.txt

2. Why SSH keys instead of a password

Passwords and keys are both ways to log in over SSH. With a password, a secret is typed and sent on every login and anyone who can reach port 22 can keep guessing. With a key pair, the **private key** (id_ed25519) never leaves the PC and the **public key** (id_ed25519.pub) is placed on the Pi. The Pi sends a challenge that only the private key can answer, so no secret crosses the network. Keys cannot be brute-forced, need no typing, and are required for scripts and automation, so password login can be switched off completely.

3. SSH keys: how they were installed and how to manage them

This section is the reusable procedure. Section 4 is the history of what happened today, including the mistakes.

3.1 Where each key lives

File	Where	What it is	Rule
id_ed25519	PC: C:\Users\Venkata\.ssh\	Private key. Proves it is you	Never share, paste, upload or commit. Back it up privately
id_ed25519.pub	PC: same folder	Public key (one line, ends chedam-dev)	Safe to copy to any machine you want to log in to

File	Where	What it is	Rule
authorized_keys	Pi: /home/chedam/.ssh/	List of public keys allowed to log in as chedam, one per line	Folder mode 700, file mode 600
ssh_host_*_key	Pi: /etc/ssh/	The Pi's own identity. New on every re-flash	Its fingerprint is remembered by the PC
known_hosts	PC: .ssh\	Fingerprints of machines already trusted	Remove a stale entry with ssh-keygen -R <ip>
config	PC: .ssh\	Shortcut so ssh chedam works	Update HostName if the Pi's IP changes

3.2 Create the key pair (once per PC)

```
ssh-keygen -t ed25519 -C "chedam-dev"          # PowerShell. Enter = default path; passphrase optional
dir $env:USERPROFILE\.ssh                     # id_ed25519 (private) + id_ed25519.pub (public)
Copy-Item $env:USERPROFILE\.ssh $env:USERPROFILE\.ssh_backup -Recurse      # backup
```

Our pair was created on 2026-10-02 and is still in use. A key is not tied to a network or an IP address, so changing Wi-Fi or router never needs a new key.

3.3 Put the public key on the Pi: method used (Raspberry Pi Imager)

- Imager → Customisation → **Remote access / SSH authentication**: enable SSH, choose **Use public key authentication**.
- Click **REMOVE** on any key already listed. Imager remembers keys from earlier sessions, and that is what broke the first attempt.
- Click **BROWSE** and select C:\Users\Venkata\.ssh\id_ed25519.pub (the .pub file, never the private key). Do not paste commands into this box.
- Check the listed key ends the same as the output of type C:\Users\Venkata\.ssh\id_ed25519.pub (here ...LuhpI0dY chedam-dev).
- Write the card. On first boot the Pi creates user chedam, writes the key into /home/chedam/.ssh/authorized_keys and turns password login over SSH off.

3.4 First connection from the PC

```
ssh-keygen -R 192.168.50.101                  # only after a re-flash: forget the old Pi identity
ssh chedam@192.168.50.101                    # type yes once; no password should be asked
ssh chedam                                    # same, via the alias in .ssh\config
```

3.5 Verify the keys on both sides

```
# PC (PowerShell): fingerprint of your public key
ssh-keygen -lf C:\Users\Venkata\.ssh\id_ed25519.pub

# Pi: the same fingerprint must appear here, plus folder/file permissions and SSH policy
ssh-keygen -lf ~/.ssh/authorized_keys
ls -ld ~/.ssh && ls -l ~/.ssh/authorized_keys      # drwx----- and -rw-----
sudo sshd -T | grep -iE "passwordauthentication|pubkeyauthentication"
# passwordauthentication no
# pubkeyauthentication yes
```

3.6 Add another key later (new laptop or phone app)

Password login is off, so a new key has to be added from a session that already works. Copy the new device's .pub file to this PC first, then append it (>> adds, it never overwrites):

```
type C:\path\to\new_device.pub | ssh chedam "cat >> ~/.ssh/authorized_keys"
```

```
ssh chedam "ssh-keygen -lf ~/.ssh/authorized_keys" # should now list 2 keys
```

3.7 Replace or remove a key

- Add the new key first (3.6) and test it in a **second** window before removing anything, so you never lock yourself out.
- Then on the Pi: `nano ~/.ssh/authorized_keys`, delete the line whose comment matches the old key (e.g. `srey . . .`), save. Changes apply to the next login, with no restart needed.
- If the PC's private key is lost or exposed, remove its line from `authorized_keys` right away and create a new pair (3.2).

3.8 If you are locked out

With password login off, the only ways back in are: a monitor and USB keyboard on the Pi (log in with the Pi password, then fix `authorized_keys`), or re-flashing the card with the correct `.pub` (3.3). Keep the `.ssh_backup` copy somewhere safe.

4. What was done, step by step

Step 1. Back up and inspect the PC's SSH folder (PowerShell)

```
Copy-Item $env:USERPROFILE\.ssh $env:USERPROFILE\.ssh_backup -Recurse
dir $env:USERPROFILE\.ssh
```

Found: `config`, `id_ed25519`, `id_ed25519.pub` (created 2026-10-02), `known_hosts`, `known_hosts.old`.

Step 2. Clear old host fingerprints

```
Remove-Item $env:USERPROFILE\.ssh\known_hosts
Remove-Item $env:USERPROFILE\.ssh\known_hosts.old
```

Iteration: the first attempt used `ssh-keygen -R <old-ip>` literally. PowerShell treats `<` as an operator, so it failed; the placeholders had to be replaced with real addresses. Deleting `known_hosts` (a backup exists) avoided needing the old IP.

Step 3. Key pair deleted by mistake, then restored

The key pair was deleted following an optional "start fresh" step, but the SD card had been written with that public key in mind, so it was restored from the backup:

```
Copy-Item $env:USERPROFILE\.ssh_backup\id_ed25519 $env:USERPROFILE\.ssh\
Copy-Item $env:USERPROFILE\.ssh_backup\id_ed25519.pub $env:USERPROFILE\.ssh\
```

Lesson: a key is not tied to a network or IP. Changing Wi-Fi never requires a new key.

Step 4. Find the Pi on the new network

The router's **DHCP Clients List** showed chedam at 192.168.50.101 (MAC 88:A2:9E:5C:17:65). `ping 192.168.50.101` replied with 0% loss.

Step 5. Login refused: "Permission denied (publickey)"

SSH was running and accepting keys only, but rejected the login. Causes found one by one:

Attempt	Result	Cause
<code>ssh 192.168.50.101</code>	denied as venkata	No user given, so SSH used the Windows username
<code>ssh chedam@chedam</code>	timed out	Windows cannot resolve the bare name chedam (use the IP or <code>chedam.local</code>)

Attempt	Result	Cause
-i \$env:venkata\.ssh\...	key file not found	\$env:USERPROFILE is a built-in variable, not a placeholder; changing it broke the path
ssh -F NUL -i C:\Users\Venkata\.ssh\id_ed25519 chedam@192.168.50.101	denied	Root cause: key mismatch. Imager had written an older saved key (...x2/COL3o srey...), while the PC holds ...LuhpI0dY chedam-dev. The old key no longer existed on the PC.

The SSH config was also stale: it pointed to the old TELUS-network address 192.168.1.241.

```
type C:\Users\Venkata\.ssh\id_ed25519.pub      # compare with the key shown in Imager
type C:\Users\Venkata\.ssh\config
# search the PC for the old key (no match found):
Get-ChildItem C:\Users\Venkata -Recurse -Filter *.pub -ErrorAction SilentlyContinue |
    Select-String "COL3o"
```

Step 6. Re-flash the SD card with the correct key

- Raspberry Pi Imager v2.0.11.1: Device Pi Zero 2 W, OS Raspberry Pi OS Lite (64-bit).
- Customisation: hostname chedam, user chedam, new Wi-Fi (2.4 GHz), country CA, SSH on with **public-key authentication**.
- **Iteration:** the PowerShell command text `Get-Content ... | Set-Clipboard` was pasted into Imager as if it were the key. Fixed by REMOVE, then BROWSE and selecting `C:\Users\Venkata\.ssh\id_ed25519.pub`.
- Write, insert the card, power on, wait about 2 minutes.

Step 7. Expected warning after re-flash, then first successful login

SSH showed *REMOTE HOST IDENTIFICATION HAS CHANGED*. This is expected: a fresh install generates a new host key. The old entry was removed and the login succeeded with no password:

```
ssh-keygen -R 192.168.50.101
ssh chedam@192.168.50.101      # type yes at the fingerprint prompt
```

Step 8. Update the PC's SSH config

```
notepad %USERPROFILE%\ssh\config

Host chedam
    HostName 192.168.50.101
    User chedam
    IdentityFile ~/.ssh/id_ed25519
```

The old `HostKeyAlias chedam.local` line was removed to avoid fingerprint confusion. `ssh chedam` now connects directly.

Step 9. Verify and update the Pi

```
ip a                                # wlan0 = 192.168.50.101, no eth0 (Zero 2 W has no Ethernet)
sudo apt update && sudo apt full-upgrade -y
sudo sshd -T | grep -i passwordauthentication    # -> passwordauthentication no
```

Step 10. Router settings (TP-Link admin at 192.168.50.1)

- **DHCP Server** → **Address Reservation:** 88:A2:9E:5C:17:65 → 192.168.50.101. It was first saved as *Disabled*; ticked and clicked **Enable Selected**.
- **IP & MAC Binding:** an entry exists (Bound), but global ARP Binding is left **Disabled**. It is a security feature, not a reservation, and can lock out unlisted devices.

Step 11. Readiness check and reboot test

```
cat /proc/device-tree/model; echo      # Raspberry Pi Zero 2 W Rev 1.0
grep PRETTY /etc/os-release           # Debian GNU/Linux 13 (trixie)
hostname -I                           # 192.168.50.101
nmcli -t -f NAME,DEVICE connection show --active # Wi-Fi on wlan0
systemctl get-default                 # multi-user.target (no desktop)
sudo reboot                           # then: ssh chedam -> works after reboot
```

Step 12. HDMI screen was blank

Diagnosis: the OS is Lite (text console only, by design; the Zero 2 W's 512 MB RAM is too little for a desktop). The kernel log showed the display driver loaded but found no screen:

```
ls /sys/class/drm/                   # card0 card0-HDMI-A-1 ...
dmesg | grep -i -E "hdmi|vc4"        # vc4-drm: [drm] Cannot find any crtc or sizes
```

Fix: force HDMI output on the kernel command line (the file must stay a single line), then reboot:

```
sudo cp /boot/firmware/cmdline.txt /boot/firmware/cmdline.txt.bak
sudo sed -i '1 s/$/ video=HDMI-A-1:1280x720@60D/' /boot/firmware/cmdline.txt
cat /boot/firmware/cmdline.txt        # ends with video=HDMI-A-1:1280x720@60D
sudo reboot
```

Result: the screen shows the chedam login: prompt. **Note:** the commands were pasted together, and the lines after `sudo cp` were eaten by the sudo password prompt, so confirm the backup exists with `ls /boot/firmware/cmdline.txt.bak`. To undo the change, remove the `video=...` text from the end of the line.

5. Lessons learned

- Run multi-line pastes one command at a time when sudo may ask for a password.
- Placeholders such as `<new-ip>` must be replaced; `$env:USERPROFILE` must not.
- Before flashing, confirm the key in Imager ends the same as type
C:\Users\Venkata\.ssh\id_ed25519.pub. Imager remembers keys from earlier sessions.
- "Permission denied (publickey)" means wrong user or a key the Pi does not hold. "Host identification has changed" after a re-flash is normal: run `ssh-keygen -R <ip>`.
- Always back up `.ssh` before changes. The backup saved the key pair today.
- With password login off, losing `id_ed25519` means re-flashing or using a keyboard and screen. Keep a safe copy.

6. Kept out of this repository

No Wi-Fi password, Pi password, router admin password or key file contents are recorded here. They belong in a password manager. Addresses listed are private LAN addresses.

7. Next steps

- Fill in the network record table in `docs/P0-restart-new-network.md` (no passwords).
- Continue P0: run `hub/scripts/setup-hub.sh` on the Pi and work through `docs/P0-readiness-checklist.md`. The script makes the hub headless (HDMI goes dark after the next reboot) unless run as `sudo bash setup-hub.sh --keep-display`.
- Remove any old "Chedam Store CA" certificate from the PC and phones, and install the new hub's CA when it is created.